

# Math camp exam

2022

Department of Economics, Columbia University

This is a timed remote exam. You have 3 hours to submit your answers via email to the TA (and cc the instructor). Total points are 100. Please justify your answers. Partial credit will be awarded in case of incomplete answers. You are not allowed to consult peers or use the web to aid in solving the problems. Columbia's academic honor code applies.

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Note:  $d, d_X, d_Y$  are used to denote generic metrics unless otherwise specified. If no metric is mentioned, assume the standard Euclidean metric  $d_2$ .

1. **(40 points, (5 points each))** Are the following true or false? Briefly justify. If false, provide a counterexample when applicable.
  - (a) In a metric space  $(X, d)$  and for  $A \subseteq X$ , the set  $A \setminus A'$  is open, where  $A'$  is the set of all limit points of  $A$ .
  - (b) In a metric space  $(X, d)$  and for  $A \subseteq X$ , if  $x \in X$  is an interior point of  $A$ , then  $x$  is a limit point of  $A$ .
  - (c) In a complete metric space  $(X, d)$  and for closed  $A \subseteq X$ , then  $A$  is a complete set.
  - (d) In the metric space  $(\mathbb{R}_{>0}, d_2)$ , the set  $(0, 1]$  is closed and bounded, and hence compact.
  - (e) Given metric spaces  $(X, d_X), (Y, d_Y)$  and function  $f : X \rightarrow Y$ , if  $f(K)$  is compact in  $(Y, d_Y)$  for any  $K$  compact in  $(X, d_X)$ , then  $f$  is a continuous function.
  - (f) In a real inner product space  $(V, +, \cdot, \langle \cdot, \cdot \rangle)$ , consider nonzero vectors  $v_1, v_2 \in V$ . If  $\langle v_1, v_2 \rangle = 0$ , then  $v_1 - v_2, v_1 + v_2$  are linearly independent.
  - (g) Consider the vector space  $(\mathbb{R}^2, +, \cdot)$  (under the usual componentwise definitions of vector addition and scalar multiplication). Then the function  $\langle \cdot, \cdot \rangle : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$  given by

$$\langle x, y \rangle = x^T A y$$

is an inner product if  $A \in \mathbb{R}^{2 \times 2}$  has all nonnegative entries.

- (h) In a real normed vector space  $(V, +, \cdot, \|\cdot\|)$  where  $V$  is a convex set, the norm is a function that is quasiconvex and homogeneous of degree 1.

2. **(8 points, (4 points each))** Consider nonempty set  $X$  having at least two elements, along with metrics on  $X$  denoted by  $d : X^2 \rightarrow \mathbb{R}_{\geq 0}, d' : X^2 \rightarrow \mathbb{R}_{\geq 0}$ . For all  $x_1, x_2, y_1, y_2 \in X$ , we define the functions

$$d_+ : X^2 \times X^2 \rightarrow \mathbb{R}_{\geq 0}, d_+((x_1, x_2), (y_1, y_2)) := \frac{1}{2}d(x_1, y_1) + \frac{1}{2}d'(x_2, y_2)$$

$$d_{\times} : X^2 \times X^2 \rightarrow \mathbb{R}_{\geq 0}, d_{\times}((x_1, x_2), (y_1, y_2)) := \sqrt{d(x_1, y_1) \cdot d'(x_2, y_2)}$$

- (a) Is  $d_+$  a metric?  
 (b) Is  $d_{\times}$  a metric?
3. **(7 points)** Suppose function  $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$  is given by

$$f(x) = \frac{1}{2} \log(1 + x).$$

- (a) **(4 points)** Is  $f$  a contraction?  
 (b) **(3 points)** Given some  $x_0 \in \mathbb{R}_{\geq 0}$ , define sequence  $(x_n)$  recursively by

$$x_{n+1} = f(x_n).$$

Does  $\lim_{n \rightarrow \infty} x_n$  exist? If so, compute it. If your answer depends on  $x_0$ , please specify how it does so.

4. **(15 points)** Consider a family of matrices  $\{A_t\}_{t \in \mathbb{R}_{\geq 0}}$ , where

$$A_t = \begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}.$$

- (a) **(8 points)** Is  $A_t$  diagonalizable in  $\mathbb{C}$ ? Is  $A_t$  diagonalizable in  $\mathbb{R}$ ? If so, diagonalize it, i.e. find a  $2 \times 2$  invertible matrix  $P_t$  and diagonal matrix  $\Lambda_t$  s.t.  $P_t^{-1} A_t P_t = \Lambda_t$ . If your answer depends on  $t \geq 0$ , please be clear on how it does so.  
 (b) **(3 points)** For what values of  $t \geq 0$  is  $A_t$  positive semidefinite? Negative semidefinite? Indefinite?  
 (c) **(4 points)** Define  $t^* := \sup \{t : A_t \text{ is positive semidefinite}\}$ . Compute  $\lim_{n \rightarrow \infty} \frac{1}{2^{n-1}} A_{t^*}^n$ .
5. **(15 points)** Consider the following function for positive integer  $n$  and real  $\alpha > 0$  :

$$F : \mathbb{R}_{>0}^{2n} \rightarrow \mathbb{R}, F(\mathbf{X}, \mathbf{Y}) = F(X_1, \dots, X_n, Y_1, \dots, Y_n) = \sum_{i=1}^n (X_i + Y_i)^\alpha.$$

- (a) **(4 points)** Is  $F$  homogeneous? If so, state its degree of homogeneity. Is  $F$  homothetic? What is

$$\left( \frac{1}{F(\mathbf{X}, \mathbf{Y})} \sum_{i=1}^n X_i \frac{\partial}{\partial X_i} F(\mathbf{X}, \mathbf{Y}) \right) + \left( \frac{1}{F(\mathbf{X}, \mathbf{Y})} \sum_{i=1}^n Y_i \frac{\partial}{\partial Y_i} F(\mathbf{X}, \mathbf{Y}) \right)?$$

- (b) **(3 points)** Is it true that for all  $\alpha \in (0, 1]$ ,  $F$  is quasiconcave? Briefly justify.

- (c) **(5 points)** Log linearize the equation

$$Z = F(\mathbf{X}, \mathbf{Y})$$

around a point  $(Z, \mathbf{X}, \mathbf{Y}) = (Z^*, \mathbf{X}^*, \mathbf{Y}^*)$  that satisfies the equation, where  $Z^*, X_i^*, Y_i^*$  are nonzero for all  $i = 1, \dots, n$ . That is, defining the log deviations<sup>1</sup>  $\hat{Z} := \log(Z/Z^*)$ ,  $\hat{X}_i := \log(X_i/X_i^*)$ ,  $\hat{Y}_i := \log(Y_i/Y_i^*)$ ,  $i = 1, \dots, n$ , use a first order Taylor approximation for  $Z - F(\mathbf{X}, \mathbf{Y})$  that is linear in  $\hat{Z}, \hat{X}_i, \hat{Y}_i$  around the point where  $\hat{Z}, \hat{X}_i, \hat{Y}_i$  are zero for all  $i = 1, \dots, n$  and set this expression equal to zero to obtain an approximated equation. You may use shortcuts discussed in class.

- (d) **(3 points)** For the case  $n = 1$ , write a second order Taylor approximation for  $F$  around the point  $(X_1, Y_1) = (a, b)$  for  $a, b > 0$ .

6. **(15 points)** Consider the following problem:

$$\max_{(x_1, x_2) \in \mathbb{R}_{\geq 0}^2} e^{\sqrt{x_1} + x_2}$$

$$\text{s.t. } x_1 + x_2 \leq m \quad (1)$$

where  $m > 0$ .

- (a) **(2 points)** Argue that a solution exists.
- (b) **(2 points)** Argue that constraint (1) must bind at any possible maximum.
- (c) **(8 points)** Solve the problem (i.e. obtain all maximizers  $(x_1, x_2)$ ).
- (d) **(3 points)** The *value function*  $v : \mathbb{R}_{>0} \rightarrow \mathbb{R}$  is given by

$$v(m) := \max_{(x_1, x_2) \in \{(a, b) \in \mathbb{R}_{\geq 0}^2 : a + b \leq m\}} e^{\sqrt{x_1} + x_2}.$$

Compute its derivative function  $v'(m)$  if it exists.

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<sup>1</sup>Alternatively, you may define percent deviations  $\hat{Z} := \frac{Z - Z^*}{Z^*}$ ,  $\hat{X}_i := \frac{X_i - X_i^*}{X_i^*}$ ,  $\hat{Y}_i := \frac{Y_i - Y_i^*}{Y_i^*}$ ,  $i = 1, \dots, n$ .